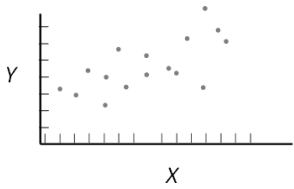
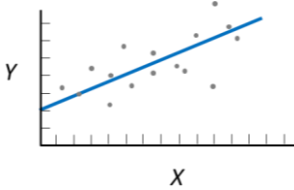


Lecture 11. Linear Regression

In Lecture 9, we learned that a **correlation** tests for an association between two continuous variables. We also learned that finding a correlation does not mean that we have proven that one variable affects the other; in other words, correlation does not imply causation.



A **linear regression** is a more complex type of analysis that also works with two continuous variables, but with a crucial advantage: it involves building a statistical **model** that describes how changes in one variable (X) correspond to changes in the other variable (Y). This makes it a highly useful tool, as it effectively enables you to **make predictions** about the value of Y (the outcome) based on a given value of X (the predictor).

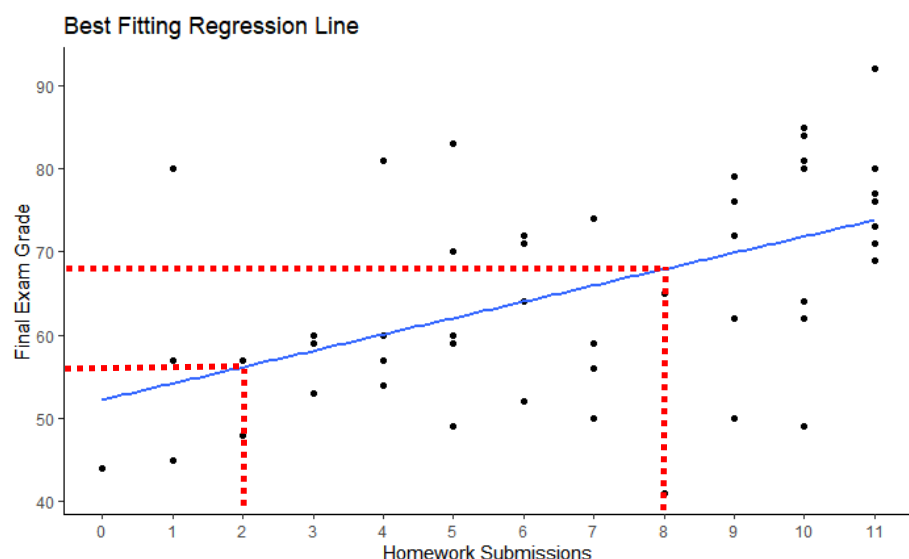


Note: The linear regression assumes that the relationship between X and Y is linear. Other types of relationships (e.g., cubic, quadratic...) will require a different type of model. For now, we'll focus only on linear relationships.

How to interpret a linear model

The linear model is visualised on graphs as a straight line going through the scatterplot.

The line represents what value we would expect Y to be when we observe a specific value of X. For example, here we have a scatterplot showing the correlation between (X) the *number of homework submissions* of different students and (Y) their *grade on the final exam*.



In this graph, we can see that submitting 2 pieces of homework (on the X-axis) corresponds to getting a score of about 56% on the exam (the Y-axis). Submitting 8 pieces of homework corresponds to getting a score of about 69% on the exam.

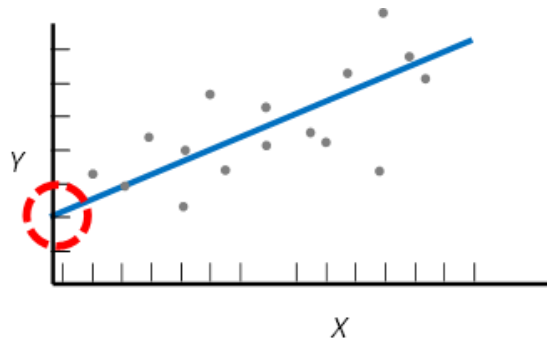
How to build a linear model?

The linear model (or regression line, or line of best fit) can be expressed mathematically using three numbers: the intercept, the slope, and the error.

Intercept

The intercept is the value of Y when $X=0$. In other words, it's the starting point of the regression line.

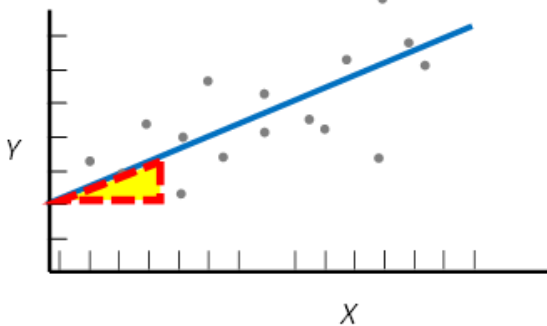
In our example above, the intercept is 52, meaning that a student who makes 0 homework submissions is expected to get around 52% on their final exam.



Slope

The slope expresses how much the value of Y changes with each degree of X . Each time X increases by 1 point, Y increases by the value of the slope.

In our example, the slope is 2, meaning that with each new homework submission your exam score increases by 2%.



We can use the intercept and the slope to predict what the outcome variable will be based on any value of the predictor we choose. To find the value of the outcome variable, we just need to start with the intercept (the starting point of the regression line) and then add the slope as many times as the value of the predictor.

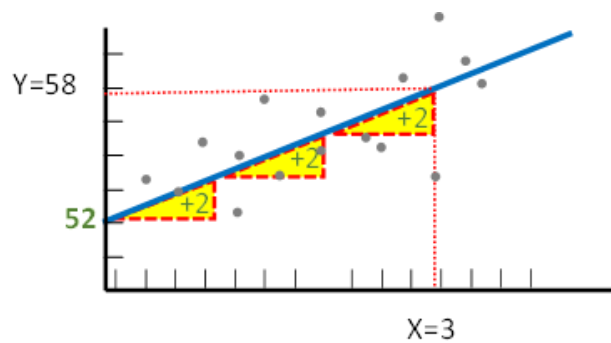
Question: In our example with the homework submissions and the exam scores, we know the intercept is 52 and a slope is 2. What exam score will a student get if she makes 3 homework submissions?

Answer: We start with the intercept of 52 – this means 0 homework submissions results in an exam score of 52%.

Adding one homework submission changes the exam score by +2 (the value of the slope), so the outcome is 54%.

Adding a second homework submissions changes it by +2 again, so the outcome is 56%.

Finally, making three homework submissions changes the score by another +2 points, resulting in a score of 58% for a student who makes 3 homework submissions.



We can express this mathematically with the following formula:

$$\text{Predicted outcome} = \text{Intercept} + (\text{Value of Predictor} \times \text{Slope})$$

Or, in mathematical notation: $Y_i = a + (b \times X_i)$

Alternative notation: $Y_i = \beta_0 + (\beta_1 \times X_i)$

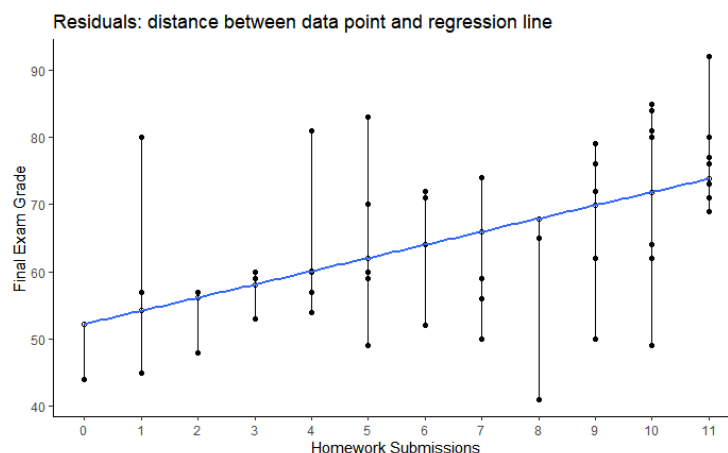
Another alternative: $y = c + (m \times x)$

Applied to our example above: $Y = 52 + (3 \times 2) = 52 + 6 = 58$

Error

Our regression line is a statistical model – a simplified description of the structure of the data. Because it is simplified, a model never describes the data perfectly, so the actually observed datapoints will deviate somewhat from the values predicted by the regression line. This discrepancy between the linear model and the real data is called the **error**.

The error is calculated through the **residuals**: the differences between the observed datapoints and the predicted values:



Testing hypotheses with regression

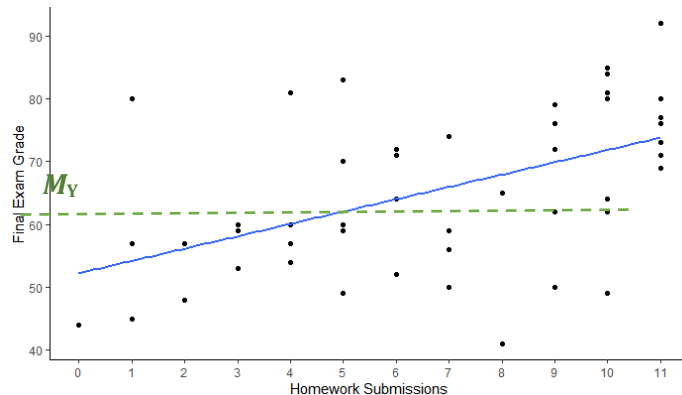
There are two types of hypothesis test for regression models:

1. **Does the model fit the data?** To test this hypothesis, we want to see whether the variance explained by the model is greater than the residual variance.
2. **Is there a significant association** between a predictor and the outcome? We test this using a t-test to see if the slope is significantly different from 0.

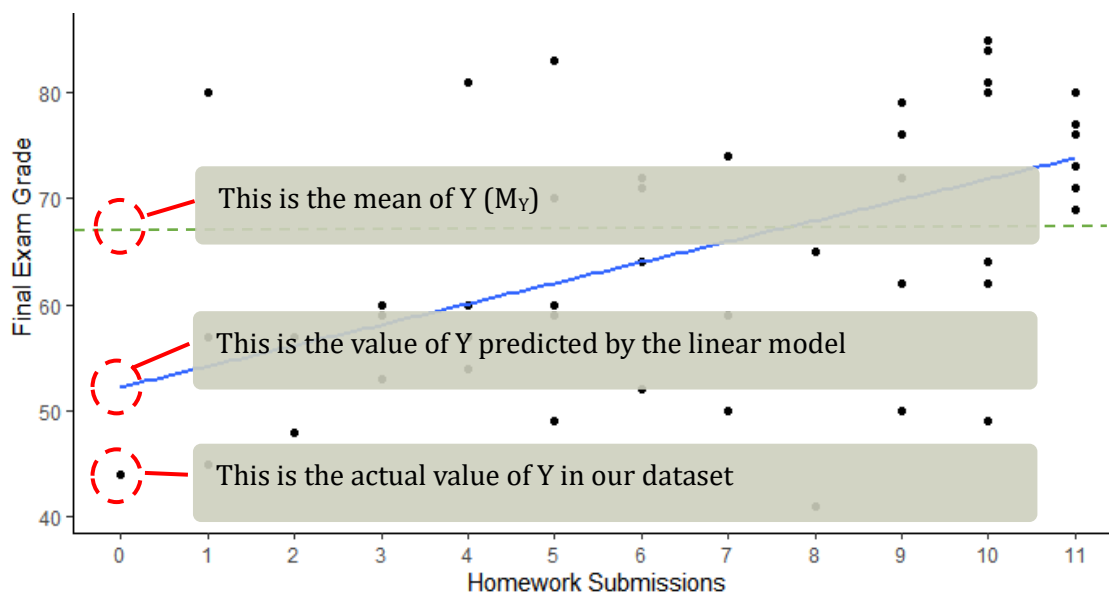
In the case of simple linear regression, these are two ways of asking the same thing and we will get the same result for both.

Hypothesis 1: Does the model fit the data?

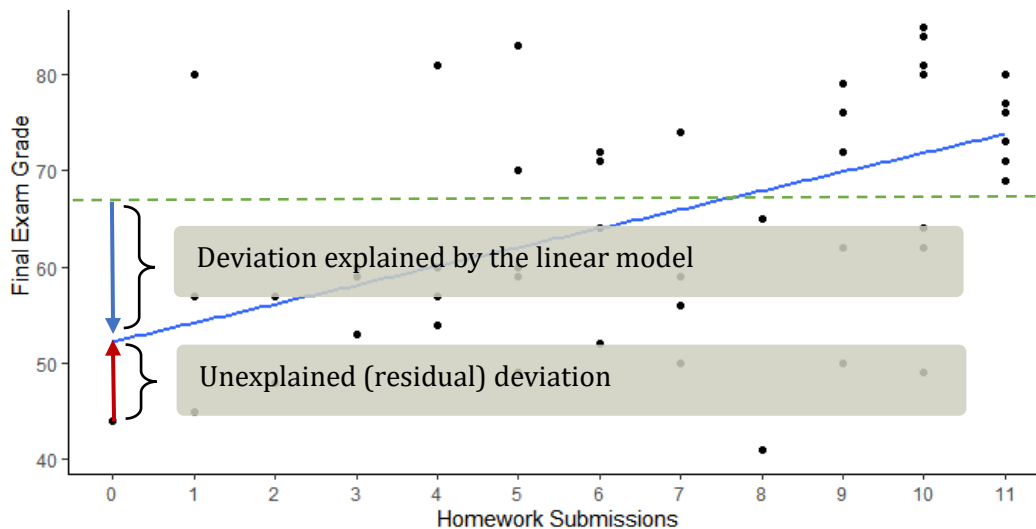
We want to test how good our model (the blue line) is at predicting variation in Y. To do that, we need to check how much each datapoint deviates from the Mean of Y (M_Y , represented here as a dotted green line), and then check how much of that deviation is explained by the regression line.



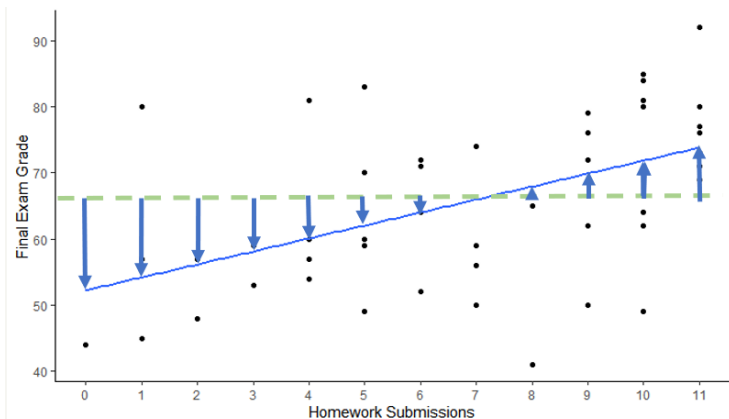
To illustrate this, let's zoom in on the first datapoint in the bottom left corner ($X=0$):



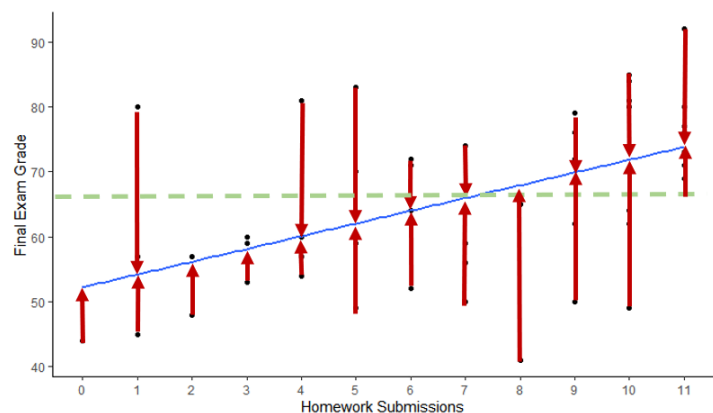
We can see that the regression line explains some of the deviation from the mean, but not all of it:



To find the **variance explained by the model**, we must calculate all the differences between M_Y and the model, square them, and add them up. This is the Sum of Squares of the Model (SSM). To get the Mean Squares of the Model (MSM), we would normally divide it by the DFs (number of predictors) of the model, but in simple linear regressions this is 1, so we'd get the same number. 



For the **unexplained variance**, we repeat the same procedure to calculate the differences between the model and the real data points, resulting in the Mean Squares of the Error (MSE). 



We can express the degree to which our model fits the data using an **F-test**, which measures the ratio of two variances:

$$F = \frac{MS_{model}}{MS_{error}}$$

Mean Squares of the Model = the variance explained by the model (deviation of predicted values from the mean value)

Mean Squares of the Error = the variance not explained (deviation of observed values from predicted values)

- H_0 : The model is no better than the mean for predicting Y
- H_1 : The model is better than the mean for predicting Y

The F value has its own **null hypothesis distribution** which is used to determine the p-value. A significant F test ($p < .05$) means the null hypothesis is rejected.

Effect size: R^2

The R^2 value is the effect size measure which tells you the proportion of the variance in Y that is accounted for by the model. For example, $R^2 = .26$ means that **26%** of the variation in exam scores is explained by the number of homework submissions – leaving another 74% of the variation in exam scores unaccounted for by the model. Obviously, the bigger the R^2 , the better X is at predicting Y.

R^2 is calculated as:

$$r^2 = \frac{SS_{model}}{SS_{total}}$$

The sum of squared differences between the predicted values and M_Y

The sum of squared differences between each observed value and M_Y

t-tests for the slope

T-tests can be used to test whether each coefficient (intercept and slope) is different from some expected value (usually 0). The intercept being different from 0 is not very informative, but the slope is. It allows us to test the null hypothesis that there is no relationship between the predictor and outcome – a significant t-test for the slope means the null hypothesis that the slope is 0 (no relationship between predictor and outcome) is rejected.

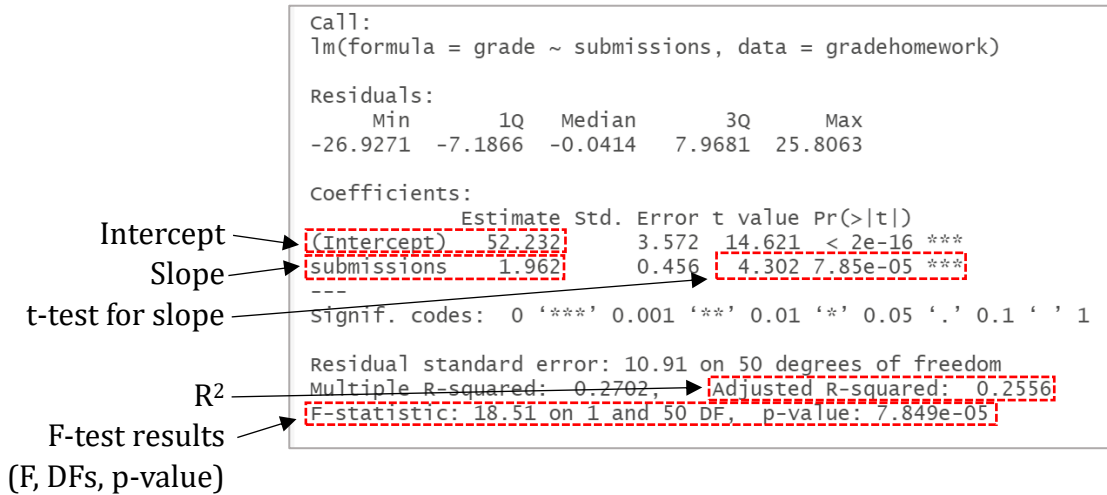
Linear Regression in R

We build the linear model using the `lm()` function, and then print the parameters of that model (intercept, slope, error, R^2 ...) with the `summary()` function:

```
# save model result
mod <- lm(grade ~ submissions, data = gradehomework)

# print summary
summary(mod)
```

The output looks like this:



```
Call:
lm(formula = grade ~ submissions, data = gradehomework)

Residuals:
    Min       1Q   Median       3Q      Max
-26.9271  -7.1866  -0.0414   7.9681  25.8063

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  52.232      3.572   14.621 < 2e-16 ***
submissions   1.962      0.456    4.302 7.85e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.91 on 50 degrees of freedom
Multiple R-squared:  0.2702,    Adjusted R-squared:  0.2556
F-statistic: 18.51 on 1 and 50 DF,  p-value: 7.849e-05
```

Intercept →

Slope →

t-test for slope →

R^2 →

F-test results (F, DFs, p-value) →

If $p < .05$, the model is a good fit for the data.

Reporting the results

We used linear regression to test the association between homework submission (predictor) and grades on the final statistics exam (outcome). The model provided better fit to the data than the null model, $F(1, 50) = 18.51$, $p < .001$. For every extra piece of homework submitted, final grades increased by 1.96 points, $t(50) = 4.302$, $p < .001$. Overall, homework submissions explained 26% of the variance in final exam scores.

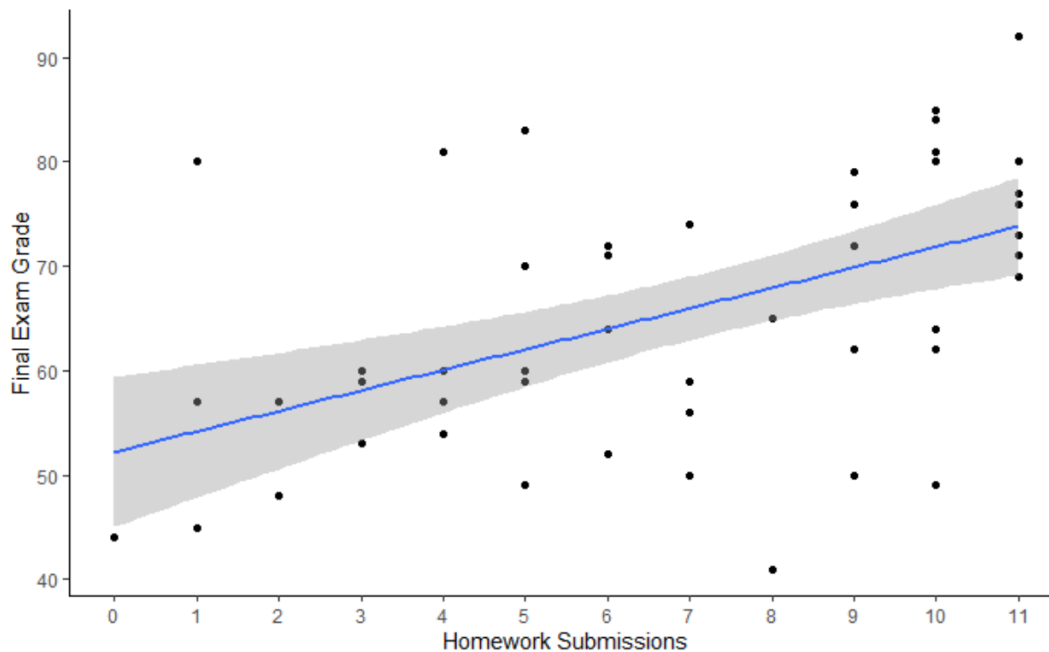
Write-up checklist:

- Report the goodness of fit of the model in comparison to the null model with the F statistic in the following format: $F(df) = f\text{-value}$, $p = p\text{-value}$.
- Describe the relationship between dependent and independent variable with reference to the theoretical constructs
- Report the slope coefficient and associated t-test result in the following format $b = \text{coefficient}$, $t(df) = t\text{value}$ $p = p\text{value}$
- Describe the coefficient in terms of the amount of change in dependent variable for each unit of change in independent variable
- Report the exact p values, unless it is $< .001$
- Report the adjusted R squared (% of variance explained)
- Report all values to 2 decimal places

Visualising the linear regression

```
show scatter plot
```{r}
ggplot(gradehomework, aes(y=grade, x=submissions)) +
 geom_point()+
 geom_smooth(method=lm)+
 theme_classic()+
 labs(y="Final Exam Grade", x="Homework Submissions")+
 scale_x_continuous(breaks=seq(0, 11, 1))
```
```

 `geom\_smooth()` using formula 'y ~ x'



Assumptions of linear regression

As with correlation:

- Interval or ratio level data
- Data points available for each participant for each of the two variables
- Linear relationship between the two variables
- Homogeneity of variance: the residuals have same distribution (and standard deviation of residuals is same for all values of X and Y)
- No extreme outliers – can distort the relationship between two variables

Plus: Residuals must be normally distributed – NOTE: you will learn how to do this in the practical